Personal Notes on Jordan Quillen (Co)homology

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Contents

Chapter 1

Background

1.1 Purpose

This chapter is a private learning layer for the mathematical background behind intrinsic Quillen homology and cohomology of Jordan algebras. It is not a claim registry and should not be treated as proof evidence.

Note 1.1. Let k be a field of characteristic 0, and let Jord_k denote a suitable category of Jordan algebras over k, for example nonunital Jordan algebras, simplicial Jordan algebras, or dg Jordan algebras over the Jordan operad.

The word *intrinsic* means that the construction is carried out inside the category of Jordan algebras itself. In particular, one does not first pass from a Jordan algebra J to an associated Lie algebra, such as the Tits–Kantor– Koecher Lie algebra, and then take Lie algebra homology.

In the nonunital absolute setting, the relevant abelianization functor is the functor of indecomposables

$$Q(J) = J/J^2,$$

where

$$J^2 := \mathrm{span}_k\{xy : x, y \in J\}.$$

The quotient J/J^2 has zero multiplication, and it is the universal zero-multiplication Jordan algebra receiving a morphism from J.

To define Quillen homology, one replaces J by a cofibrant replacement

$$\widetilde{J} \xrightarrow{\sim} J$$

in the chosen homotopical category of Jordan algebras, and then applies Q. Thus the derived indecomposables are

$$\mathbb{L}Q(J) := Q(\widetilde{J}) = \widetilde{J}/(\widetilde{J})^2.$$

The intrinsic Jordan Quillen homology of J is

$$H_n^{\mathrm{Jord}}(J) := H_n\big(\mathbb{L}Q(J)\big).$$

More generally, for a morphism of Jordan algebras

$$A \longrightarrow B,$$

one expects a relative cotangent complex

$$\mathbb{L}^{\mathrm{Jord}}_{B/A}.$$

If M is an appropriate B-module, for example a Beck module or Jordan module, then the intrinsic Jordan Quillen cohomology is defined by

$$D^n_{\mathrm{Jord}}(B/A, M) := H^n \operatorname{Hom}_B\big(\mathbb{L}^{\mathrm{Jord}}_{B/A}, M\big).$$

Dually, one may define homology with coefficients by

$$D^{\mathrm{Jord}}_n(B/A, M) := H_n\big(\mathbb{L}^{\mathrm{Jord}}_{B/A} \otimes_B M\big),$$

or, more carefully, by using the appropriate derived tensor product.

The low-degree interpretation should be analogous to the commutative and Lie cases:

$$D^0_{\mathrm{Jord}}(B/A, M) \cong \operatorname{Der}^{\mathrm{Jord}}_A(B, M),$$

while

$$D^1_{\mathrm{Jord}}(B/A, M)$$

should classify square-zero extensions of B by M, subject to the precise choice of the coefficient category.

1.2 Working Setting

The default ground field is a field k of characteristic 0. The current working category is usually nonunital Jordan algebras, with augmented unital algebras treated through their augmentation ideals when that convention is appropriate.

Definition 1.1. A Jordan algebra is a commutative algebra whose product satisfies the Jordan identity. In this project the product is written as xy in mathematical prose and sometimes as $x * y$ in code-oriented notes.

1.3 Indecomposables

The guiding abelianization candidate is

$$Q(J) = J/J^2,$$

where J^2 denotes the linear span of products xy. The slogan is useful only after the ambient category and the unital convention have been fixed.

Question 1.1. What is the cleanest first setting for $Q(J) = J/J^2$: absolute nonunital Jordan algebras, augmented unital Jordan algebras, or a relative category?

Note 1.2. In the absolute nonunital category, the abelian objects are precisely the Jordan algebras with zero multiplication. Indeed, a vector space V with product

$$uv = 0 \qquad \text{for all } u, v \in V$$

is automatically a Jordan algebra. Therefore the natural abelianization of a nonunital Jordan algebra J is not a commutator quotient, but rather

$$Q(J) = J/J^2.$$

Here J^2 is the linear span of all products. The quotient has zero multiplication because every product in J/J^2 is represented by an element of J^2.

Let A be a unital Jordan algebra over k. An *augmentation* of A is a unital Jordan algebra morphism

$$\varepsilon : A \longrightarrow k,$$

where k is regarded as a unital Jordan algebra with its usual multiplication.

The pair (A, ε) is called an *augmented unital Jordan algebra.* Its *augmentation ideal* is

$$A_+ := \ker(\varepsilon).$$

Since $\varepsilon(1_A) = 1$, the canonical inclusion

$$k \longrightarrow A, \qquad \lambda \longmapsto \lambda 1_A$$

splits ε as a map of k-vector spaces. Hence there is a vector space decomposition

$$A \cong k \cdot 1_A \oplus A_+.$$

Moreover, A_+ is an ideal. Indeed, if $a \in A_+$ and $b \in A$, then

$$\varepsilon(ab) = \varepsilon(a)\varepsilon(b) = 0,$$

so $ab \in A_+$. In particular, if $a, b \in A_+$, then $ab \in A_+$. Thus A_+ is naturally a nonunital Jordan algebra.

The augmentation ideal is important because, for a unital algebra A, the naive quotient

$$A/A^2$$

is usually useless: since $1_A \in A$, one typically has

$$A^2 = A.$$

Therefore, in the augmented unital setting, the correct analogue of indecomposables is

$$Q(A) := A_+/A_+^2,$$

where

$$A_+ = \ker(\varepsilon).$$

This is the same idea as in the nonunital setting, but applied to the augmentation ideal rather than to the whole unital algebra.

Conversely, if J is a nonunital Jordan algebra, one may form its unitalization

$$J^+ := k \oplus J.$$

The product is defined by

$$(\lambda, x)(\mu, y) := \big(\lambda\mu, \lambda y + \mu x + xy\big),$$

where $\lambda, \mu \in k$ and $x, y \in J$. The element

$$(1, 0)$$

is the unit. There is a natural augmentation

$$\varepsilon : J^+ \longrightarrow k, \qquad \varepsilon(\lambda, x) = \lambda,$$

and its kernel is

$$\ker(\varepsilon) = 0 \oplus J \cong J.$$

Thus nonunital Jordan algebras may often be studied equivalently through augmented unital Jordan algebras by passing between

$$J \qquad \text{and} \qquad k \oplus J.$$

1.4 Modules, Extensions, And Differentials

The coefficient objects should be understood through square-zero extensions $J \ltimes M$, Beck modules over J, and Jordan derivations. The main research question is whether the Beck-module convention exactly matches the classical Jordan-module convention used in the literature. (I need to ask my supervisor about this).

The expected Jordan analogue of Kahler differentials is a universal object $\Omega_{B/A}$ representing Jordan derivations. The corresponding cotangent complex $L^{\mathrm{Jord}}_{B/A}$ is still provisional until the resolution framework is fixed.

Chapter 2

Workspace Architecture

2.1 Repository Map

The repository separates mathematical claims, computational evidence, source code, paper text, and private notes. This separation is part of the research method: a computation may suggest a claim, but it does not by itself prove it.

2.2 Theory And Claims

`theory/` stores the evolving mathematical development. The directory `theory/claims/` is the formal claim registry, and every theorem-like statement should eventually be represented there with status, assumptions, dependencies, proof sketch, references, and verification notes.

The current claim statuses are deliberately conservative. `CLAIM-0001` and `CLAIM-0002` are draft claims, while `CLAIM-0003` through `CLAIM-0005` are conjectural.

2.3 Code, Tests, And Experiments

`src/jordan_qh/` contains exact Python scaffolding for small examples, linear algebra, identities, derivations, modules, omega candidates, indecomposables, and toy Quillen checks. The matching tests live in `tests/`.

`experiments/` records planned or executed computations. The experiment numbers are stable workspace identifiers, not a logical dependency order.

2.4 Paper, Formalization, And Private Notes

`paper/` is the public-facing LaTeX skeleton and should only receive statements that are aligned with checked or clearly labelled claim statuses. `formal/` is a Lean scaffold, not part of the default verification path.

`note/` is a private learning area. This file may contain bilingual explanations, partial understanding, and personal TODOs, but it should not silently promote experiments or conjectures into proved results.

Chapter 3

Experiments

3.1 Reading Rule

Each experiment should be read as computational evidence with an explicit boundary. A result file may report a successful run, but the corresponding mathematical claim still needs independent review.

3.2 Experiment 001: Square-Zero Sanity Checks

Experiment 001 has status *run.* It checks one-dimensional split square-zero extensions $E = J \ltimes M$ with exact rational arithmetic.

The result is a sanity check for scalar actions in the idempotent and zero-multiplication one-dimensional cases. It does not prove the Beck-module classification and does not change the status of `CLAIM-0003`.

3.3 Experiment 002: Two-Dimensional Jordan Algebras

Experiment 002 has status *not run yet.* Its goal is to enumerate or import small two-dimensional Jordan algebra examples and test derivations, indecomposables, and candidate differential modules.

3.4 Experiment 003: Beck Module Identities

Experiment 003 has status *not run yet.* It is intended to symbolically derive the identities imposed on M by the Jordan identity on a square-zero extension $J \ltimes M$.

3.5 Experiment 004: TKK Comparison

Experiment 004 has status *not run yet.* Its purpose is to compare intrinsic Jordan Quillen invariants with Lie homology of the associated TKK Lie algebra on small examples.

3.6 Experiment 005: Derived Indecomposables Toy Benchmark

Experiment 005 has status *run.* It benchmarks a one-generator toy model for derived indecomposables on $F(x)$, $F(x)/(x^2 - x)$, $F(x)/(x^2)$, $F(x)/(x^3)$, and $F(x)/(x^4)$.

The rule is that after applying Q, only the linear part of a relation survives. The current generated result says that all five planned examples match their expected H_0 and H_1 dimensions.

3.7 Experiment 006: Presentation-Invariance Stress Test For J_3

Experiment 006 has status *run*. It studies

$$J_3 = (t)/(t^3)$$

through Presentation A, $F(x)/(x^3)$, and a naive diagnostic for Presentation B, $F(x,y)/(y - x^2, xy, y^2)$.

The current result computes $H_0 = 1, H_1 = 1$ for Presentation A. The naive one-step diagnostic for Presentation B computes $H_0 = 1, H_1 = 2$, exposing an extra false class caused by the incomplete relation complex.

3.8 Experiment 007: Two-Step Cofibrant Replacement Toy For J_3

Experiment 007 has status *run*. It keeps Presentation B from Experiment 006,

$$F(x,y)/(y - x^2, xy, y^2),$$

and adds one explicit degree 2 generator b to kill the extra naive class represented by a_3.

The hand-coded two-step replacement uses degree 0 generators x, y, degree 1 generators a_1, a_2, a_3, and differential

$$\begin{aligned} d(a_1) &= y - x^2, \\ d(a_2) &= xy, \\ d(a_3) &= y^2, \\ d(b) &= a_3 - xa_2 - (y + x^2)a_1 + x(xa_1). \end{aligned}$$

After applying indecomposables $Q(-) = (-)/(-)^2$, the complex is

$$Q_2 = \langle b \rangle \longrightarrow Q_1 = \langle a_1, a_2, a_3 \rangle \longrightarrow Q_0 = \langle x, y \rangle$$

with

$$d_Q(a_1) = y, \qquad d_Q(a_2) = 0, \qquad d_Q(a_3) = 0, \qquad d_Q(b) = a_3.$$

The generated result records $d_Q^2 = 0$, and the symbolic check behind the degree 2 attachment reduces in this fixed one-generated example to

$$x^2x^2 - x(xx^2) = 0,$$

using power-associativity in the subalgebra generated by x.

The computed homology dimensions are

$$H_0 = 1, \qquad H_1 = 1, \qquad H_2 = 0,$$

with representatives x in H_0, a_2 in H_1, and no H_2 representative. Compared with Experiment 006, the naive Presentation B H_1-dimension drops from 2 to 1.

The boundary is essential: this verifies the fixed low-degree two-step repair for J_3 Presentation B through degree 1. It is not an automatic syzygy search, not a general cofibrant replacement algorithm, and not a proof of full presentation invariance.

Chapter 4

Experiment Summary

4.1 What The Current Runs Support

The current executed experiments support the usefulness of exact arithmetic small examples. Experiment 001 checks square-zero extension behavior in tiny families, Experiment 005 validates a one-generator toy rule, and Experiment 006 shows why presentation issues cannot be ignored. Experiment 007 then repairs the fixed J_3 Presentation B diagnostic through degree 1 by adding one explicit degree 2 generator.

4.2 What The Current Runs Do Not Prove

The successful toy runs do not prove the full Jordan Quillen homology theory. They do not establish a general cofibrant replacement construction, a presentation-invariance theorem, or an equivalence between intrinsic Jordan Quillen homology and TKK Lie homology.

4.3 Most Important Boundary

The most important current warning is the pair of Experiments 006 and 007. Experiment 006 shows that a naive one-step relation complex for Presentation B has an extra false H_1 class. Experiment 007 shows that this particular false class can be killed by a hand-coded degree 2 generator, but the repair is still example-specific and low-degree.

4.4 Personal Takeaway

The experiments are best used as guardrails for definitions. They are useful when they expose impossible claims, missing hypotheses, or misleading computational shortcuts. The 007 run is especially useful as a template for what later automatic low-degree cell attachment would need to reproduce.

Chapter 5

Theory Roadmap

5.1 Abelianization

The first theoretical target is to make $Q(J) = J/J^2$ precise in the chosen category. The related claim files are currently draft-level and should not be quoted as checked theorems.

5.2 Beck Modules And Square-Zero Extensions

The next target is to decide whether Beck modules over a Jordan algebra match the classical Jordan module identities. This is central because it controls the coefficient category for cohomology.

5.3 Derivations And Universal Differentials

After the coefficient category is clear, the project should construct and test universal Jordan differentials $\Omega_{B/A}$. The universal property remains a conjectural or proof-draft target until the module convention is settled.

5.4 Cotangent Complex And Low Degrees

The cotangent complex $L^{\mathrm{Jord}}_{B/A}$ requires a resolution framework, such as simplicial Jordan algebras, dg Jordan algebras over the Jordan operad, or a comparison between both.

Experiment 007 suggests the next concrete computational target: replace naive one-step relation complexes by a principled low-degree cell-attachment procedure. The present evidence is only the fixed J_3 two-step repair, so it should guide implementation tests rather than be quoted as a theorem.

Low-degree cohomology should eventually explain D^0 in terms of Jordan derivations and D^1 in terms of square-zero extensions. These statements need separate claim files before they become paper-level assertions.

5.5 Comparison Problems

The comparison with Glassman cohomology and TKK Lie-theoretic constructions should remain explicit and cautious. Agreement in small examples is useful, but non-agreement may be equally informative.

Chapter 6

Current Repository Assessment

6.1 Snapshot Boundary

This chapter records a working assessment of the repository state. It is a private planning note, not a claim file and not proof evidence. Any statement below that depends on generated results should be checked against the relevant `results.json`, `experiment.md`, commit record, and verification commands before it is quoted in public project documentation.

Note 6.1. The current project stage is best described as

$$\text{reproducible experimental scaffold} \quad + \quad \text{small exact computations.}$$

It is not yet a completed theory of intrinsic Jordan Quillen homology. The useful achievement is narrower but important: the repository now has enough notation, claims, tests, and toy computations to expose false shortcuts and to shape the next definitions.

6.2 Overall Diagnosis

The project goal is clear. The intended construction is intrinsic: use the Jordan algebra category itself, rather than first replacing a Jordan algebra by an associated Lie algebra such as the Tits–Kantor–Koecher algebra. The guiding objects are derived abelianization, Jordan cotangent complexes, low-degree Quillen cohomology, square-zero extensions, and comparisons with Glassman or TKK-style constructions.

The repository discipline is also good. It asks for a strict distinction between proved claims, proof drafts, conjectures, and computations. It also requires exact arithmetic where possible and forbids fabricated experiment output. This is especially important here because small Jordan examples can look convincing even when a naive relation complex is not homotopically invariant.

At the code level, the package is intentionally light: a Python package over Python 3.11 or later, with no runtime dependencies and development tools such as `pytest`, `ruff`, and `pre-commit`. The `Makefile` exposes entries such as `test`, `lint`, `claims`, and `experiments`. A current practical caveat is that `makeexperiments` does not mean that every experiment is rerun: the top-level `scripts/run_all_experiments.py` currently behaves as a conservative status/reporting surface and prints that experiments are not run by default.

6.3 Claim Status

The theory layer remains deliberately conservative.

- `CLAIM-0001` is draft-level: zero-multiplication Jordan algebras are the expected abelian objects in the relevant nonunital setting.
- `CLAIM-0002` is draft-level: the indecomposables functor $Q(J) = J/J^2$ is expected to be the abelianization or left adjoint in the chosen absolute nonunital setting.
- `CLAIM-0003` is conjectural: Beck modules over Jordan algebras should match a precise Jordan module convention, but the identity extraction still needs to be completed.
- `CLAIM-0004` is conjectural: universal Jordan differentials $\Omega_{B/A}$ need a settled coefficient category and universal property.
- `CLAIM-0005` is conjectural: the transitivity triangle for the Jordan cotangent complex is not yet proved in the project.

6.4 Experiment Status Matrix

The current experiment landscape is:

001 square-zero, run.
A finite-grid sanity check for one-dimensional split extensions.

002 two-dimensional, not run.
Representative small examples still need to be computed.

003 Beck modules, not run.
Symbolic module identities are still a placeholder target.

004 TKK comparison, not run.
Comparison examples have not yet been computed.

005 derived Q, run.
The one-generator toy benchmark matches the expected dimensions.

006 J_3 stress test, run.
Naive Presentation B has an extra false H_1 class.

007 two-step repair, run.
One hand-coded degree 2 generator kills that false class in the tested low-degree complex.

The status should be read from each experiment's own files, especially `results.json` or `results.md`. The global runner is not yet a complete experiment execution engine.

6.5 Experiment 001: Split Square-Zero Extensions

Experiment 001 studies one-dimensional split square-zero extensions

$$E = J \ltimes M, \qquad M^2 = 0.$$

It tests two one-dimensional bases:

$$e^2 = e \qquad \text{and} \qquad e^2 = 0,$$

with $M = km$ and scalar action

$$L_e(m) = \lambda m.$$

The important methodological point is that the experiment uses exact rational arithmetic and checks the Jordan identity on a finite grid of linear combinations, not only on basis vectors. This catches a real failure mode: $\lambda = 2$ can pass a basis-only smoke test but fail once general linear combinations are tested.

For $e^2 = e$, the tested nonunital split extension passes for

$$\lambda \in \{0, \tfrac{1}{2}, 1\}.$$

If $(e, 0)$ is required to remain a unit in the extension, the surviving value is $\lambda = 1$. For $e^2 = 0$, the tested action is forced to

$$\lambda = 0.$$

This experiment supports the square-zero workflow behind `CLAIM-0003`, but it does not prove the Beck-module classification. Its main lesson is that future module-identity checks must use symbolic derivation or a strong polynomial identity check, not basis-vector testing alone.

6.6 Experiment 005: One-Generator Derived Indecomposables

Experiment 005 is the cleanest current Quillen-style toy computation. It studies one-generator presentations

$$F(x), \quad F(x)/(x^2 - x), \quad F(x)/(x^2), \quad F(x)/(x^3), \quad F(x)/(x^4).$$

The toy rule is that after applying indecomposables Q, only the linear part of a relation enters the first relation complex. In other words, the current toy complex is the two-term complex

$$k^{\{\text{relations}\}} \longrightarrow k^{\{\text{generators}\}},$$

where the differential records relation linear terms. Exact row-rank computations then give H_0 and H_1.

The current benchmark conclusions are:

presentation	(H_0, H_1)
$F(x)$	$(1, 0)$
$F(x)/(x^2 - x)$	$(0, 0)$
$F(x)/(x^2)$	$(1, 1)$
$F(x)/(x^3)$	$(1, 1)$
$F(x)/(x^4)$	$(1, 1)$.

This gives a useful low-degree intuition for $Q(J) = J/J^2$: a relation with a linear term can kill H_0, while a pure higher-order relation appears as a naive H_1 class in this one-step model. The boundary remains essential: this is not a general cofibrant replacement engine and it does not establish presentation invariance.

6.7 Experiment 006: Presentation Stress Test For J_3

Experiment 006 compares two presentations of

$$J_3 = (t)/(t^3).$$

The first presentation is

$$\text{A: } F(x)/(x^3),$$

and the second is

$$\text{B: } F(x,y)/(y-x^2,\ xy,\ y^2).$$

Presentation A agrees with the one-generator toy computation and gives

$$H_0 = 1, \qquad H_1 = 1.$$

The naive one-step diagnostic for Presentation B gives

$$H_0 = 1, \qquad H_1 = 2.$$

The result is therefore not a failure of the project. It is a useful negative test: the naive relation complex sees an extra false H_1 class for Presentation B. Without higher syzygy data or a genuine cofibrant replacement, Presentation B cannot be used as evidence for presentation invariance.

Note 6.2. The practical target created by Experiment 006 is precise: identify the extra class in the naive H_1, then attach a higher cell or syzygy generator that kills it without changing the intended algebra.

6.8 Experiment 007: Two-Step Toy Repair For J_3

Experiment 007 performs exactly that fixed repair. It starts from Presentation B and adds one degree 2 generator b, designed to kill the redundant class represented after applying Q by a_3, the relation corresponding to $y^2 = 0$.

The low-degree indecomposable complex has the form

$$Q_2 = \langle b \rangle \longrightarrow Q_1 = \langle a_1, a_2, a_3 \rangle \longrightarrow Q_0 = \langle x, y \rangle,$$

with

$$d_Q(a_1) = y, \qquad d_Q(a_2) = 0, \qquad d_Q(a_3) = 0, \qquad d_Q(b) = a_3.$$

Hence b kills exactly the extra class that was visible in the naive Presentation B computation.

The computed dimensions become

$$H_0 = 1, \qquad H_1 = 1, \qquad H_2 = 0.$$

Thus the naive H_1-dimension 2 from Experiment 006 drops to 1, in agreement with Presentation A through the tested degree.

This is the most important current mathematical evidence in the repository: the project has moved from detecting a presentation artifact to hand-coding a low-degree semi-free repair that removes it. The result must still be quoted carefully. It is a fixed J_3 computation, not an automatic syzygy search, not a general cofibrant replacement algorithm, and not a proof of full presentation invariance.

6.9 Current Strengths

The main strengths of the current repository are:

- The research boundary is explicit: computation is evidence, not proof.
- The claim files are conservative and do not overstate the theory.
- The package already has exact linear algebra utilities for small examples.
- The indecomposables code computes the rank of J^2 and therefore $\dim Q(J) = \dim J - \dim J^2$ in finite examples.

- The tests cover basic cases such as zero multiplication, a one-dimensional idempotent algebra, and a one-dimensional square-zero algebra.
- Experiments 006 and 007 give a concrete warning and repair pattern for presentation-sensitive toy computations.

6.10 Current Risks

The main risks are:

1. The coefficient theory is not yet settled. The files for modules, universal differentials, and TKK comparison remain draft or placeholder surfaces rather than completed theory.
2. Experiment 007 is hand-coded. It should inspire the next algorithm, but it is not itself an automatic cell-attachment or syzygy-discovery procedure.
3. Experiments 002, 003, and 004 are still the missing bridge between the one-generator toy computations and the broader Jordan algebra theory.
4. The global experiment runner does not yet summarize the whole experiment state in a machine-checkable matrix.

6.11 Next Experimental Priorities

Priority 1: Experiment 008

The next best experiment is an automatic syzygy-discovery version of Experiment 007. The goal is to input the three first-stage relations

$$a_1, \quad a_2, \quad a_3$$

from Presentation B and automatically find the degree 2 attachment whose image under Q kills a_3. The expected output should reproduce

$$H_0 = 1, \qquad H_1 = 1, \qquad H_2 = 0,$$

while explicitly saying that this is only a degree 1/2 toy replacement, not a full cofibrant replacement.

Priority 2: Experiment 003

The Beck-module identity experiment should be restored and made symbolic. The starting point should be the Jordan identity

$$((a^2b)a) = a^2(ba)$$

inside a square-zero extension $J \ltimes M$ with $M^2 = 0$. The experiment should extract the M-linear terms and compare the resulting operator identities for L_x with the classical Jordan module axioms. This is central for `CLAIM-0003`.

Priority 3: Experiment 002

The two-dimensional example table should be completed in a modest, representative way before attempting classification. Good first examples are: the two-dimensional zero product algebra, $ke \oplus km$ with $e^2 = e$, a one-dimensional idempotent plus a zero ideal, and a small two-dimensional nilpotent example. For each example, record $\dim J^2$, $\dim Q(J)$, derivation dimensions, and candidate relations for Ω.

Priority 4: Experiment 004

The TKK comparison should begin with two or three minimal examples and should allow disagreement as a valid outcome. The repository rule is correct: one should not identify intrinsic Jordan Quillen homology with TKK Lie homology without either a proof or a counterexample.

Priority 5: Experiment Status Aggregator

The script `scripts/run_all_experiments.py` should become a true read-only status aggregator before it becomes an execution driver. A useful first version would read each `experiment.md` and `results.json` or `results.md`, then produce a matrix recording status, pass/fail data, and the presentation-invariance flags that already appear in the JSON outputs. By default it should avoid rewriting results.

6.12 Working Conclusion

The most important current mathematical progression is

$$\text{Experiment 006} \quad \longrightarrow \quad \text{Experiment 007.}$$

Experiment 006 shows that a naive Presentation B relation complex for J_3 contains an extra false H_1 class. Experiment 007 shows that a specific two-step semi-free toy replacement kills that class and restores the expected low-degree dimensions through degree 1.

The next phase should therefore not jump directly to paper-level theorems. It should first turn the hand-coded success of Experiment 007 into a repeatable low-degree method, while also completing the coefficient-theory and small example experiments that make the theory meaningful.

Appendix A

Notation And Literature

A.1 Notation Anchor

The canonical notation table is `theory/latex_notation.md`. This private note should follow that file instead of inventing competing notation.

A.2 Basic Notation

The most important current notations are: ground field k, Jordan algebra J, module M, product span J^2, indecomposables $Q(J) = J/J^2$, cotangent complex $L^{\mathrm{Jord}}_{B/A}$, homology $D^{\mathrm{Jord}}_n(B/A, M)$, and cohomology $D^n_{\mathrm{Jord}}(B/A, M)$.

A.3 Literature Notes

The curated literature notes live in `literature/notes/`. They are the Codex-facing layer for references; local PDFs remain under `literature/papers-local/` and should be checked by the human reader when precise formulas or citations matter.

A.4 Verification Reminders

For Python changes, the default checks are:

- `python-mpytest`
- `python-mruffcheck.`
- `pythonscripts/check_claims.py`

For this private note, the relevant check is that the LaTeX file compiles.